# SCEMENT: SCALABLE AND MEMORY EFFICIENT INTEGRATION OF LARGE-SCALE SINGLE CELL RNA-SEQ DATA

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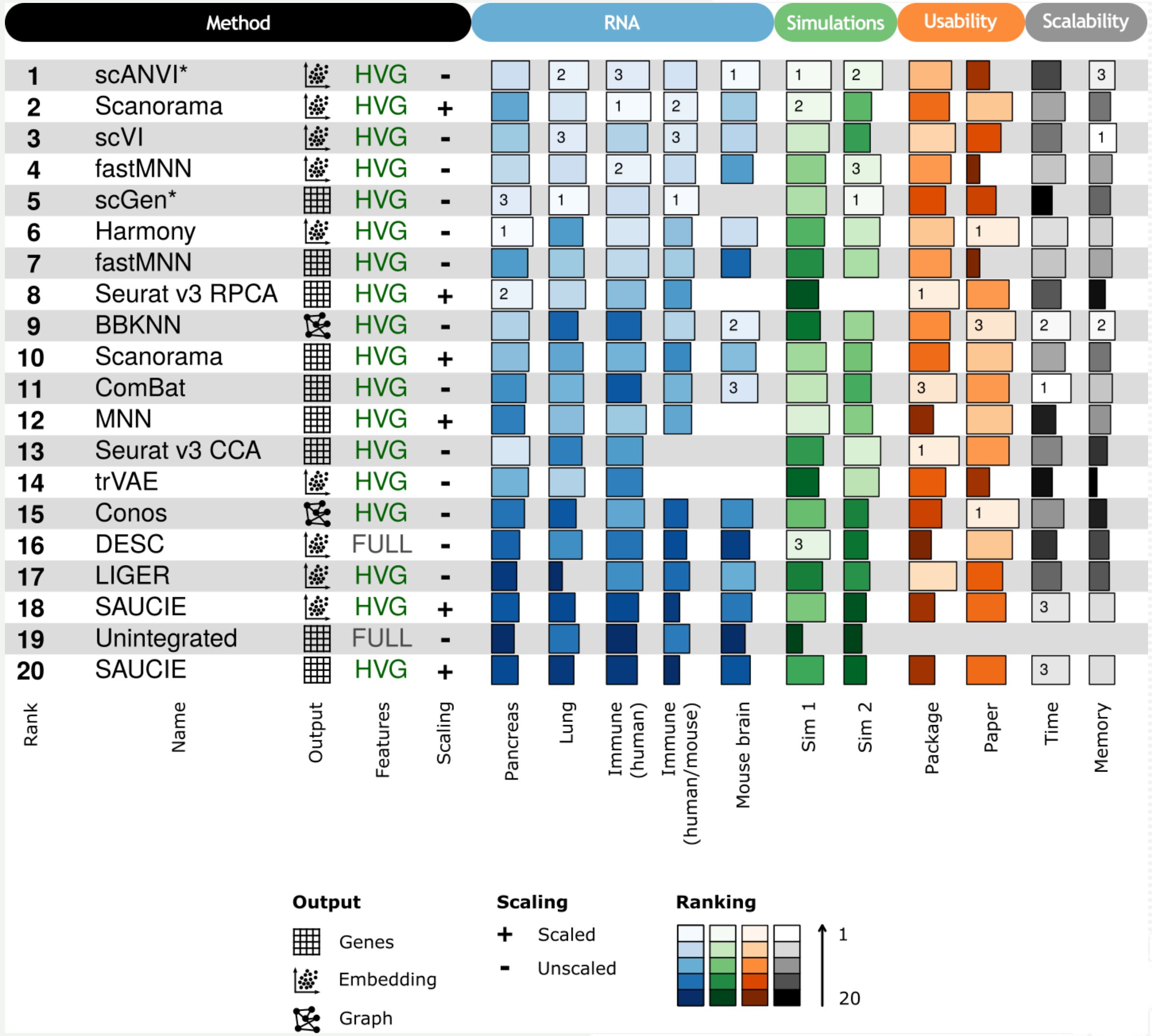
# Large-scale single cell RNA-seq Integration

- Large number of single cell datasets available in public repositories
- Integration of multiple single cell datasets promises improved understanding of biological complexity.
- Lack of methods to integrate large-scale datasets due to memory and runtime constraints.
- Primary Goal
  - Given a large number of single-cell datasets, eliminate batch effects while conserving biological variation
  - Retain quantitative gene expression information in the integrated data

# Lueken *et. al.* (2022) survey



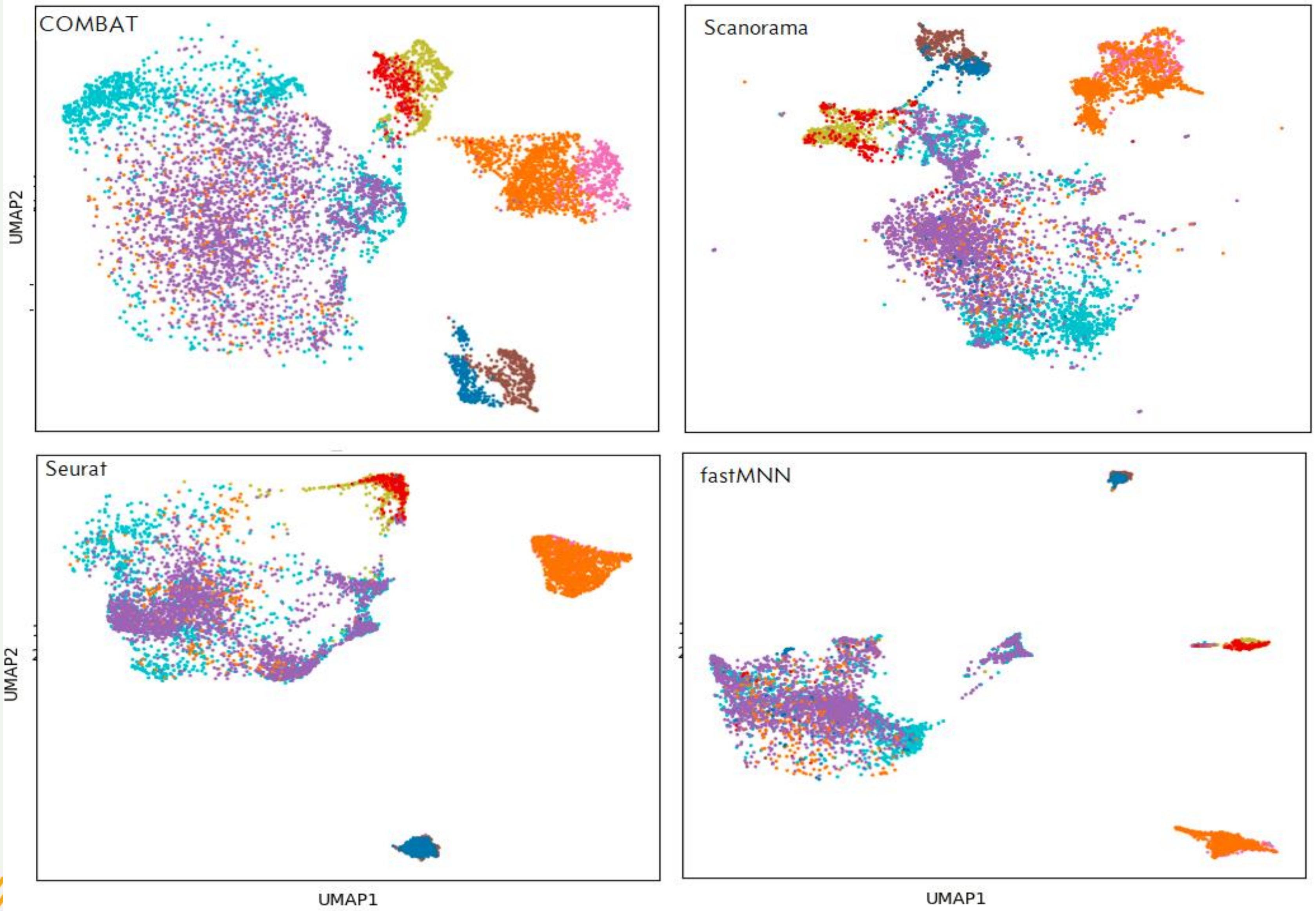

- Survey of 19 different integration methods for scRNA-seq using 14 different quality metrics.
- Unsupervised methods that retain gene-level quantitative Information
  - fastMNN
  - Seurat v3
  - Scanorama
  - ComBat
- For a mouse brain dataset of 970K cells, none of the above methods successfully completed their run.

# Comparison of Methods

| | COMBAT | Scanorama | Seurat v3 | fastMNN |
|---|---|---|---|---|
| Methodology | Linear Model | Panorama Stitching | CCA | MNN |
| No. of Genes | Union | Union | Intersection | Intersection |
| Scalability | 100K cells for limited #genes | ~1M cells for limited #genes | ~120K cells for limited #genes | Limited # genes |
| Speed (seconds)* | 91 | 113 | 2526 | 188 |
| Memory (GB)* | 9 | 5 | 55 | 5 |

*Speed & Memory on 2 Arabidopsis Datasets of 8 batches with total of 10,579 cells

# scRNA-seq Mouse Aortic Valve Dataset




17,985 cell Dataset

(Xu et.al. Arteriosclerosis, Thrombosis, and Vascular Biology. 2020. *Cell-Type Transcriptome Atlas of Human Aortic Valves Reveal Cell Heterogeneity and Endothelial to Mesenchymal Transition Involved in Calcific Aortic Valve Disease.*)

Cell Type Annotation:

- Diseased-Lymphocytes
- Healthy-Lymphocytes
- Diseased-Macrophages
- Healthy-Macrophages
- Diseased-VEC
- Healthy-VEC
- Diseased-VIC
- Healthy-VIC

# SCEMENT Model

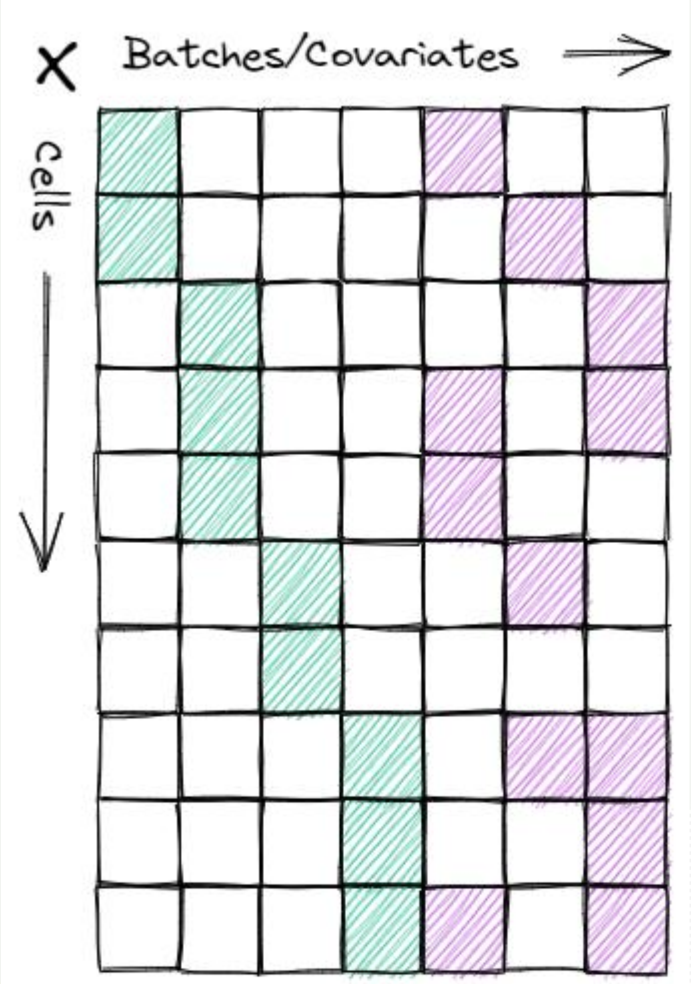
Linear Model for gene expression:

$$Y_{ijg} = \alpha_g + X_j\beta_g + \gamma_{ig} + \delta_{ig}\varepsilon_{ijg}$$

*where,*

| | |
|---|---|
| $Y_{ijg}$ | Gene expression of gene ‘g’ in cell ‘j’ of batch ‘i’ |
| $\alpha_g$ | Overall expression of gene ‘g’ |
| $X_j$ | Vector of variables that influence gene expression in cell ‘j’ (called covariates)<br>• Covariates can be either categorical variables or numerical variables |
| $\beta_g$ | Vector of regression coefficients corresponding to gene ‘g’ |
| $\gamma_{ig}$ | Additive batch effects |
| $\delta_{ig}$ | Multiplicative batch effects |
| $\varepsilon_{ijg}$ | Error term, assumed to follow Normal distribution |
| $X$ | Design matrix (N Cells x B Covariates)<br>• Covariates include batch information, experimental conditions, genotypes, etc. |

# SCEMENT: Computing $\hat{\beta}$ via Regression




$$\hat{\beta} = (X^T X)^{-1} X^T Y^T$$

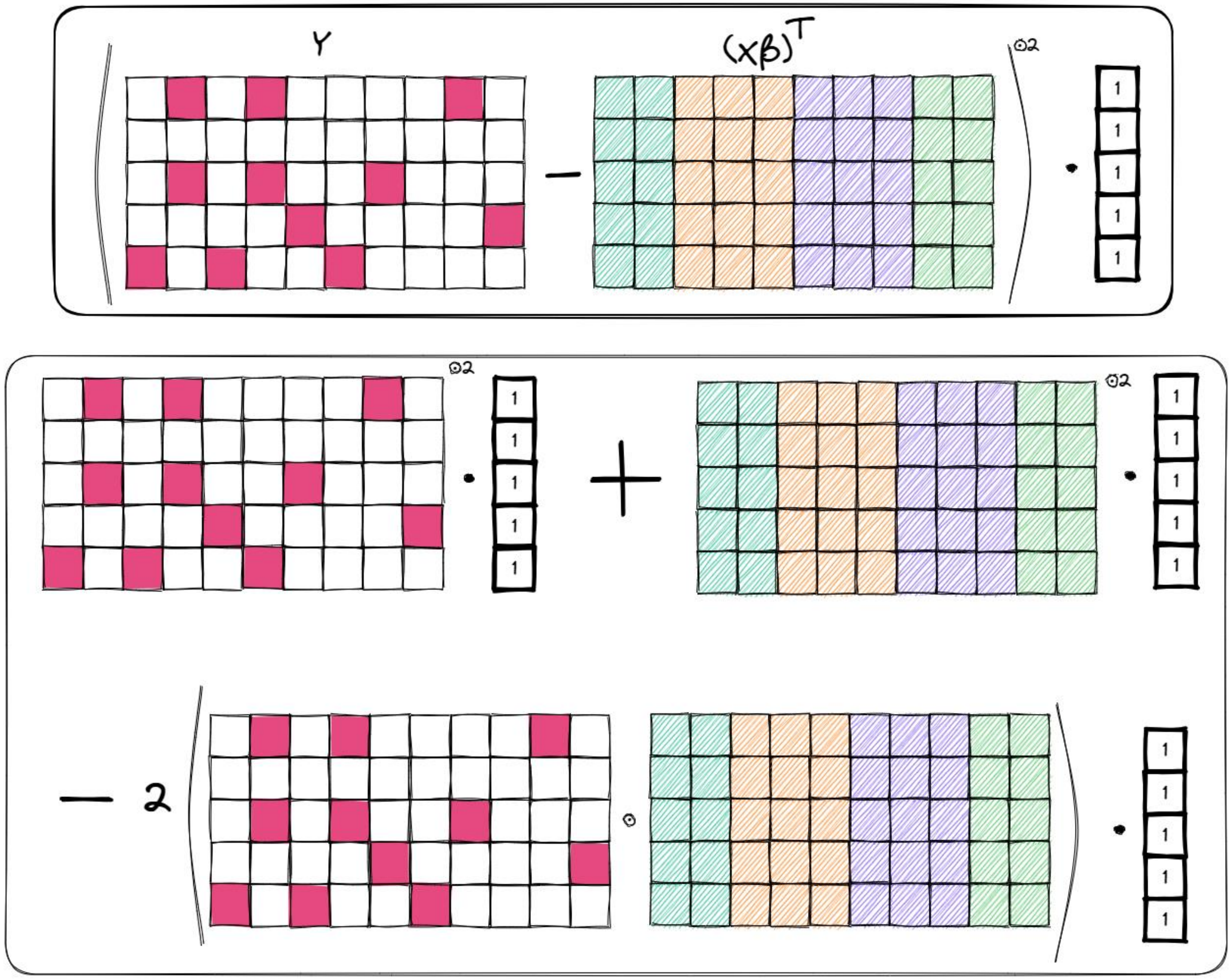
- $(X^T X)^{-1} X^T Y^T$ can be computed either
  $[(X^T X)^{-1} X^T] Y^T$ (OR) $(X^T X)^{-1} [X^T Y^T]$
- Excluding time taken for $(X^T X)^{-1}$ ,
  1. the first way : $B^2 N + BNG$ time
  2. the second way : $BNG + B^2 G$ time
- In the case when
  – N is in the order of millions,
  – G in the order of few thousands,
  – The second way is faster than the first way

# SCEMENT: Computing $\hat{\sigma}$

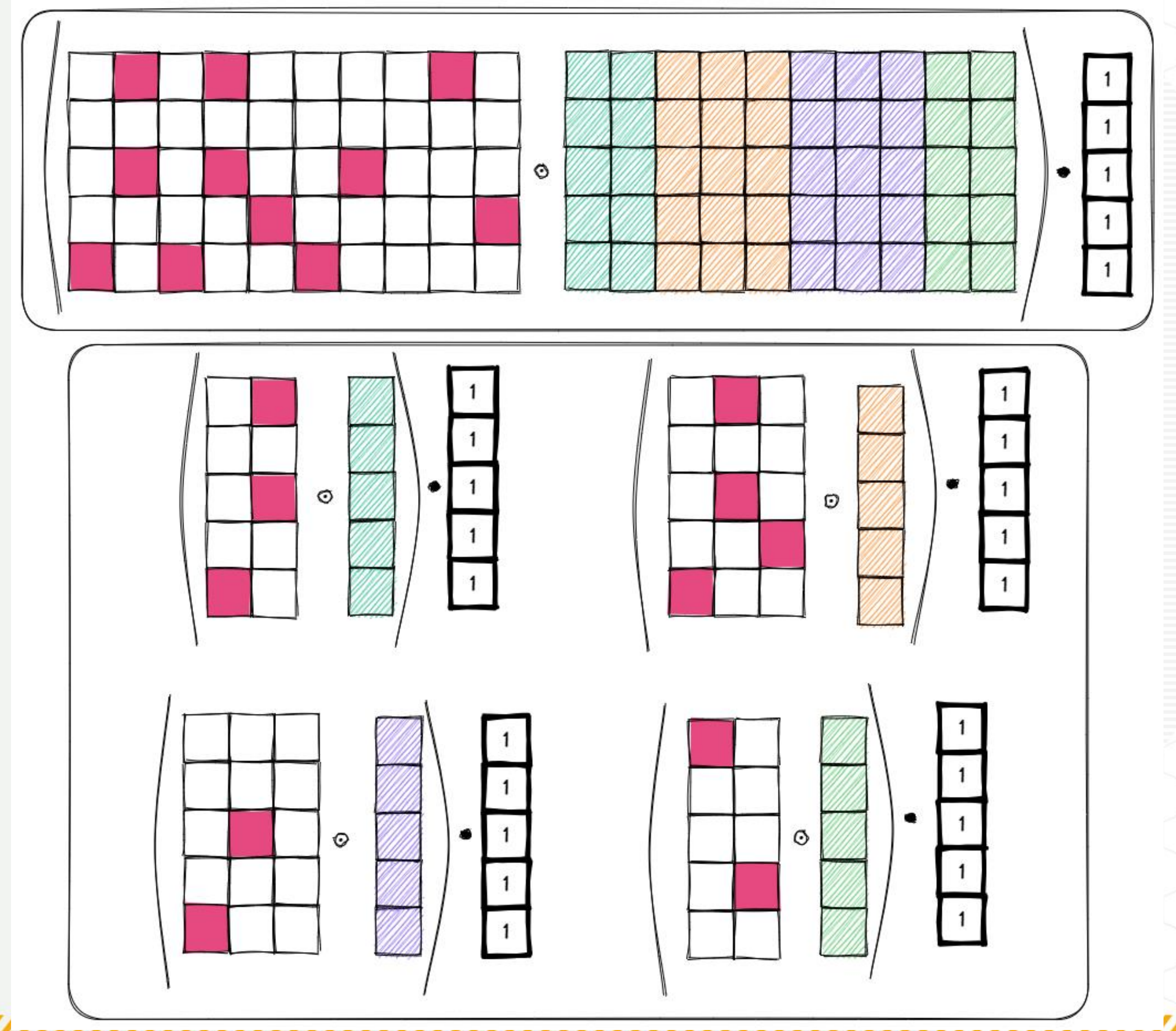
$$\left(Y - (X\hat{\beta})^T\right)^{\circ 2} \cdot 1_N^T$$

- Entries in X are repetitive.
- Expand $\left(Y - (X\hat{\beta})^T\right)^{\circ 2}$ as sum of three vectors
  - $Y^{\circ 2} \cdot 1_N^T$
  - $\left((X\hat{\beta})^T\right)^{\circ 2} \cdot 1_N^T$
  - $(2Y \circ (X\hat{\beta})^T) \cdot 1_N^T$

    (From $(a+b)^2 = a^2 + b^2 + 2ab$)



# SCEMENT: Computing $\hat{\sigma}$



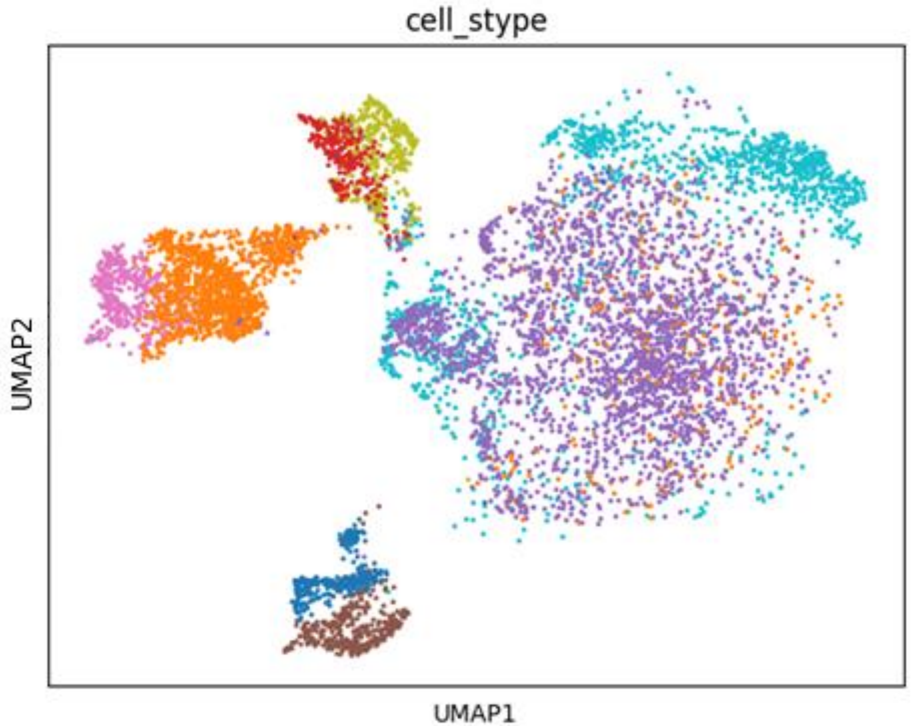

Computing $\left(Y - (X\hat{\beta})^T\right)^{\circ 2} \cdot 1_N^T$

- Expanding $(2Y \circ (X\hat{\beta})^T) \cdot 1_N^T$
  (From $(a+b)^2 = a^2 + b^2 + 2ab$)

- Similar computations occur in computing $\widehat{\gamma_{ig}}$ and $\widehat{\delta_{ig}}$.

# Implementation

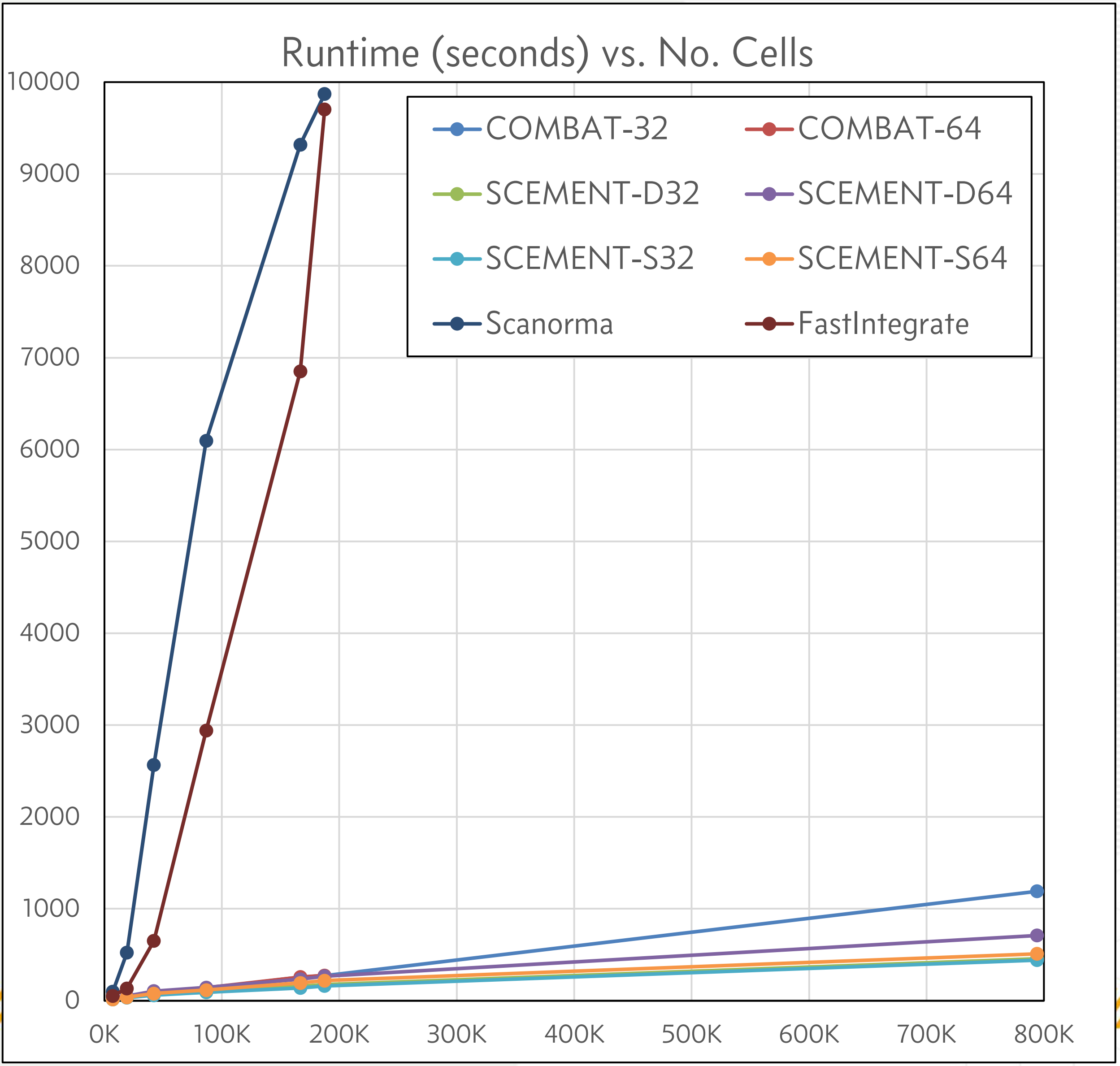
- Python implementation
  - Sparse and Dense options
  - Single precision (32-bit floating pt.) and double precision (64-bit floating pt.) options
- Dense implementation uses numpy arrays
- Sparse implementation uses scipy sparse libraries
- Construction of Design matrix
  - Using "formulaic" python library
  - Matrix of uint8 to save space
- Same Results : as shown in Aortic Datasets UMAP

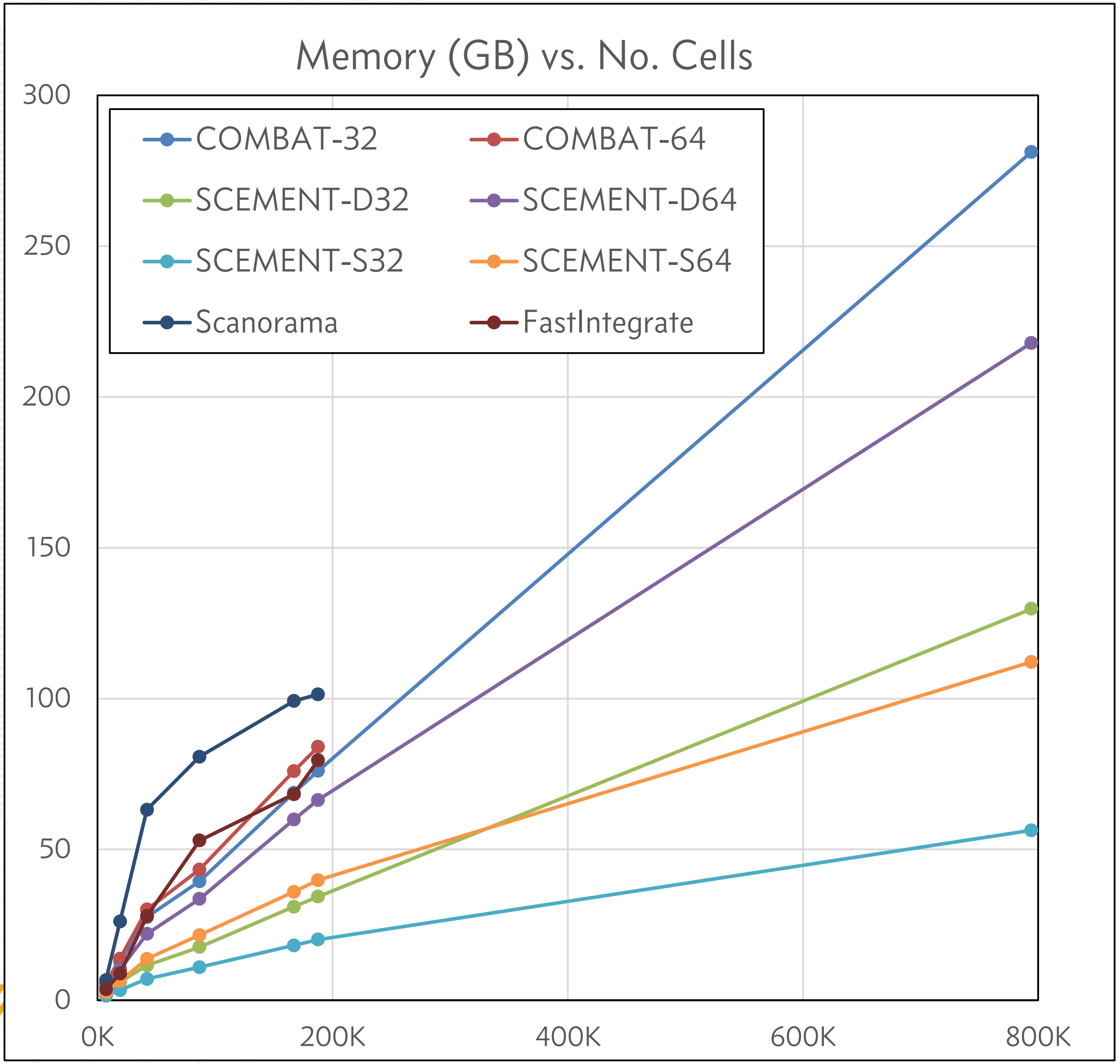


# Scalability evaluation with PBMC datasets

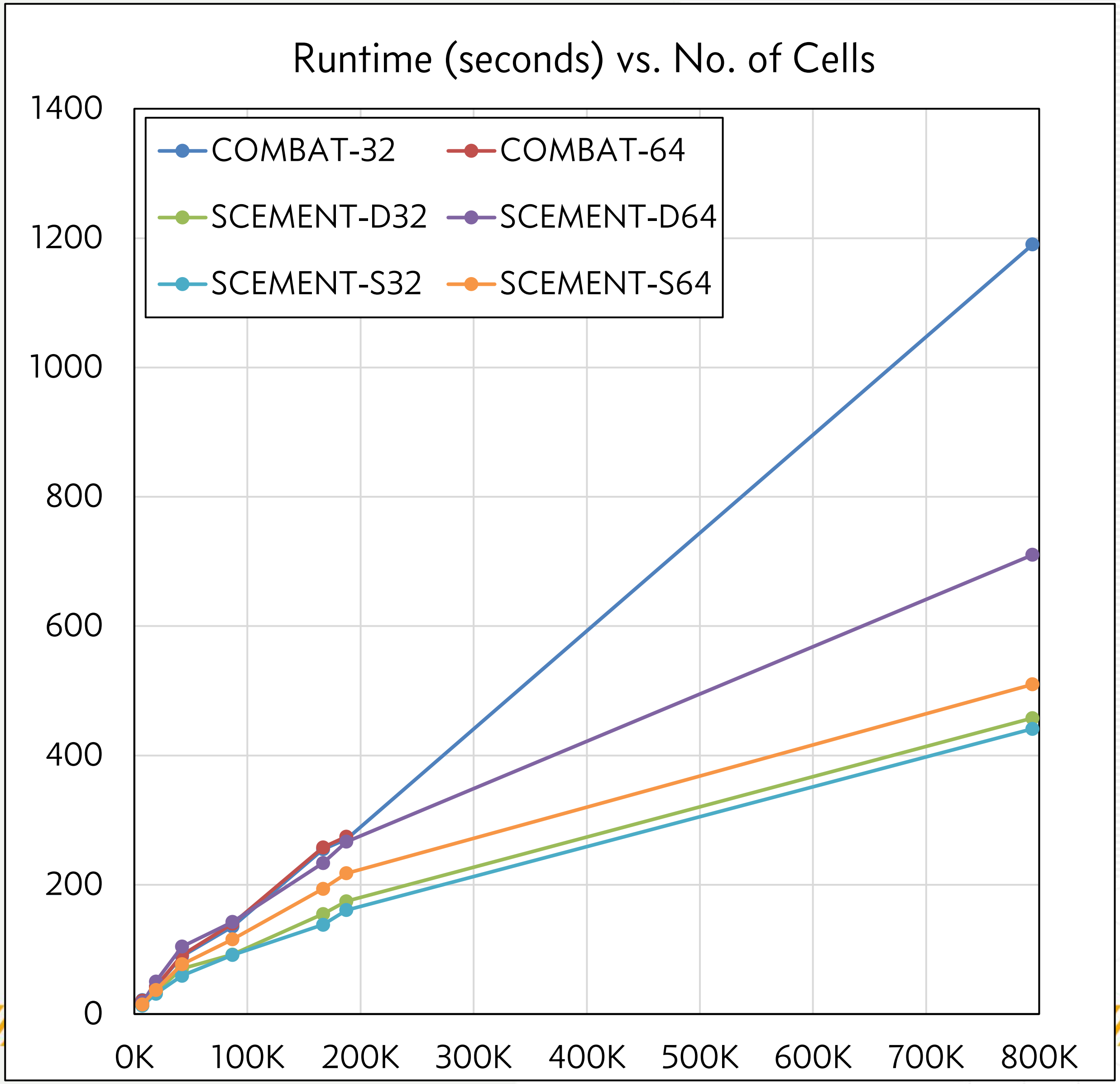
| No. of Datasets | No. of Cells | No. Genes after Intersection | No. of Genes after Union |
|---|---|---|---|
| 2 | 7,245 | 17,092 | 18,741 |
| 3 | 19,167 | 16,824 | 22,869 |
| 5 | 42,132 | 16,700 | 23,982 |
| 9 | 86,692 | 12,337 | 25,621 |
| 13 | 166,932 | 11,602 | 25,980 |
| 16 | 187,564 | 11,451 | 26,390 |
| 17 | 794,170 | 8,432 | 34,177 |

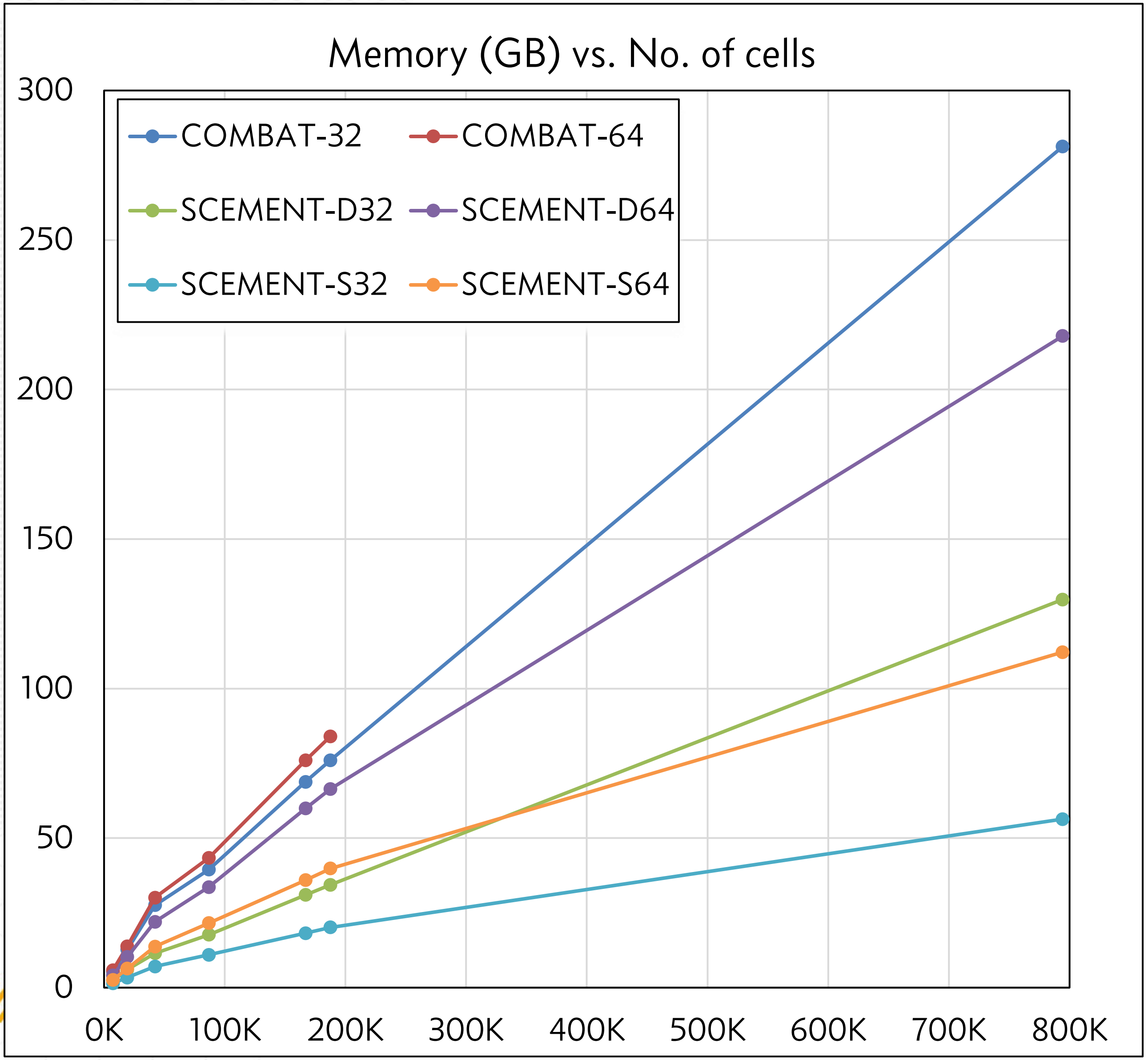
# Scalability w. Intersection of Genes



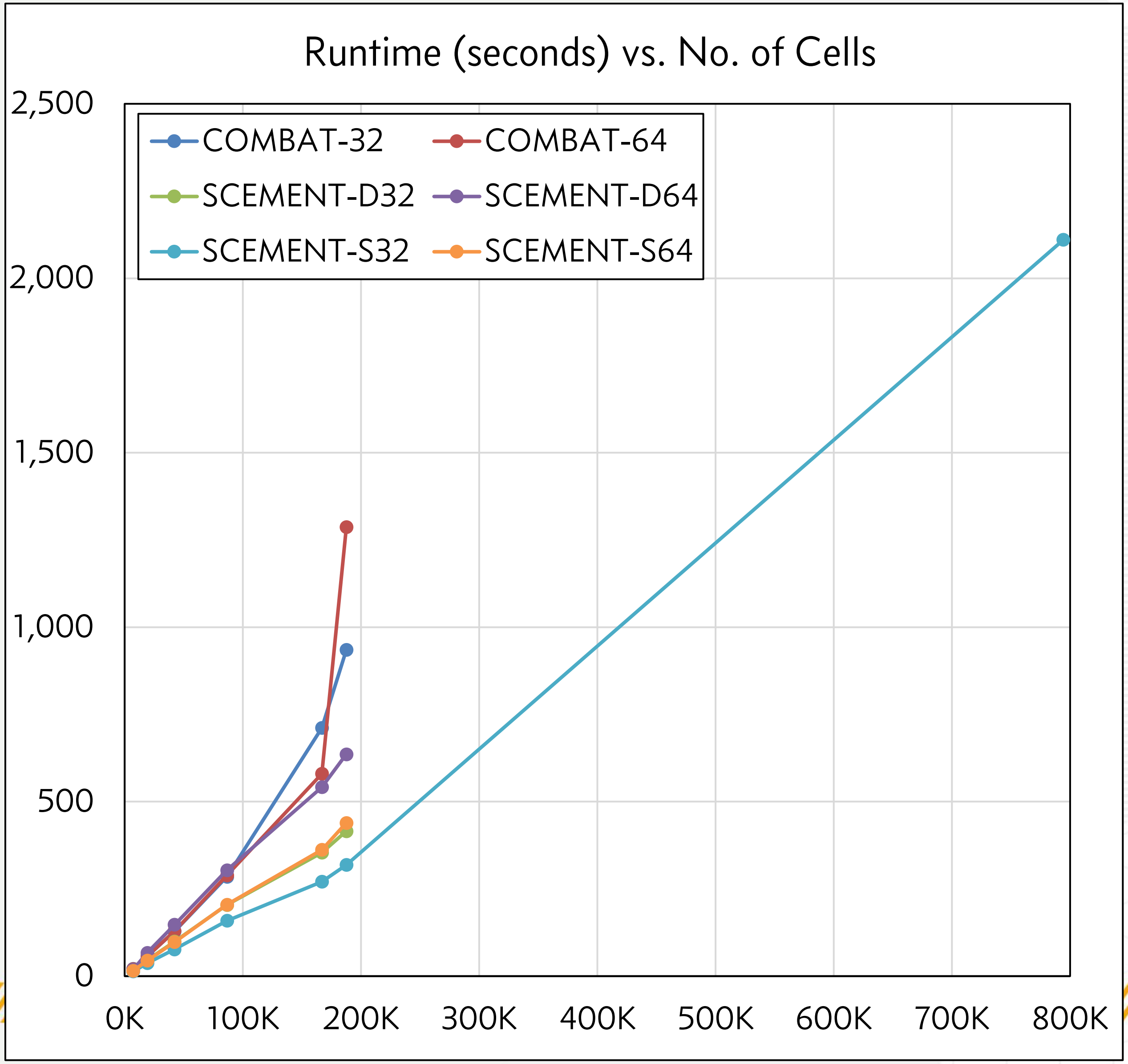



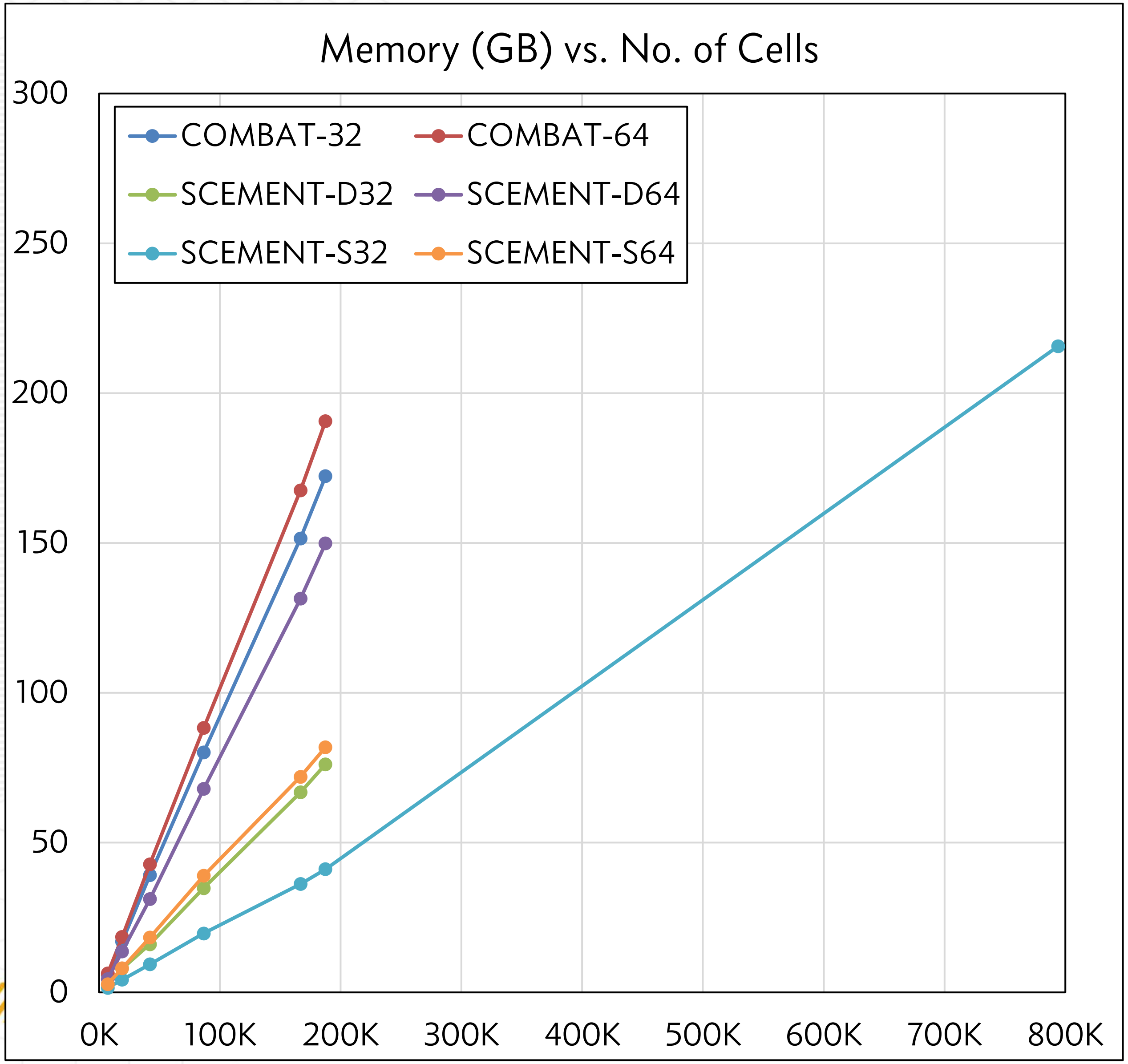
# Scalability w. Intersection of Genes



# Scalability w. Union of Genes



# **descartes** Human Gene Expression during Development

- Ref:
  - Cao, J. et al. A human cell atlas of fetal gene expression. Science 370, eaba7721 (2020).
  - https://descartes.brotmanbaty.org/bbi/human-gene-expression-during-development/
- 4,062,980 cells
- 15 organs and 77 cell types
- 2000 Highly Variable Genes

| Phase | Runtime (minutes) |
|---|---|
| Pre-processing Data | 77.25 |
| Integration | 46.01 |
| Clustering Workflow (including UMAP) | >48 hours (Didn’t finish) |

# Conclusions

- SCEMENT is a memory-efficient integration method for scRNA-seq datasets.
- Time and Memory efficient for <200K cells datasets
  - Compared to Scanorama and FastIntegrate, upto ~6X reduced memory  and upto ~50X faster.
  - Compared to COMBAT, upto ~6X reduced memory  and upto ~3X faster.
- Makes large-scale data integration feasible.
- Future Directions:
  - Parallel algorithm to improve speed.
  - Faster algorithm for UMAP and clustering.

# Acknowledgements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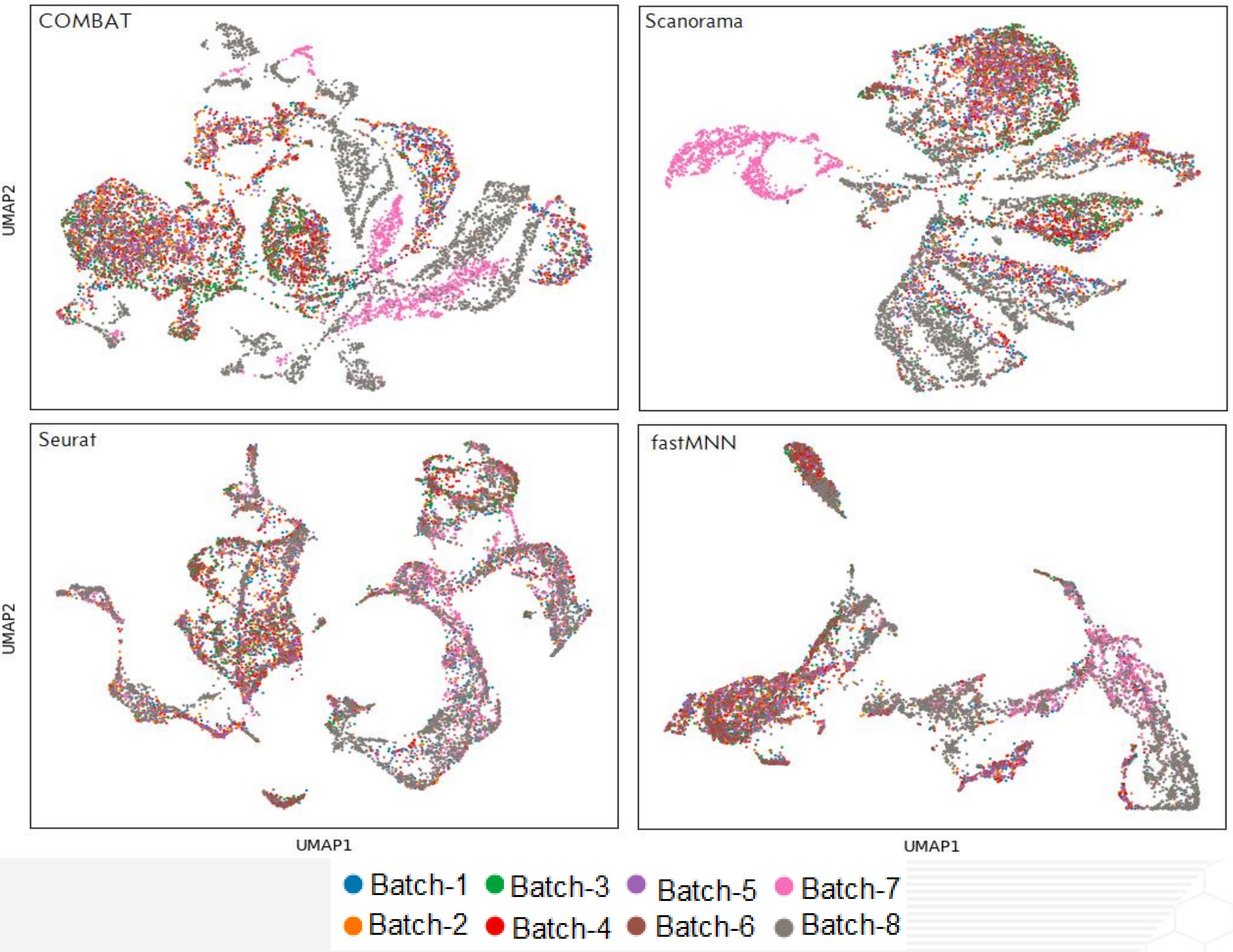
# SCIB Metrics for two *Arabidopsis thaliana* datasets

- GSE158761
  - 6658 cells
  - 6 batches
  - 3 time points
- GSE121619
  - 3921 cells
  - 2 batches
  - With and without heat stress

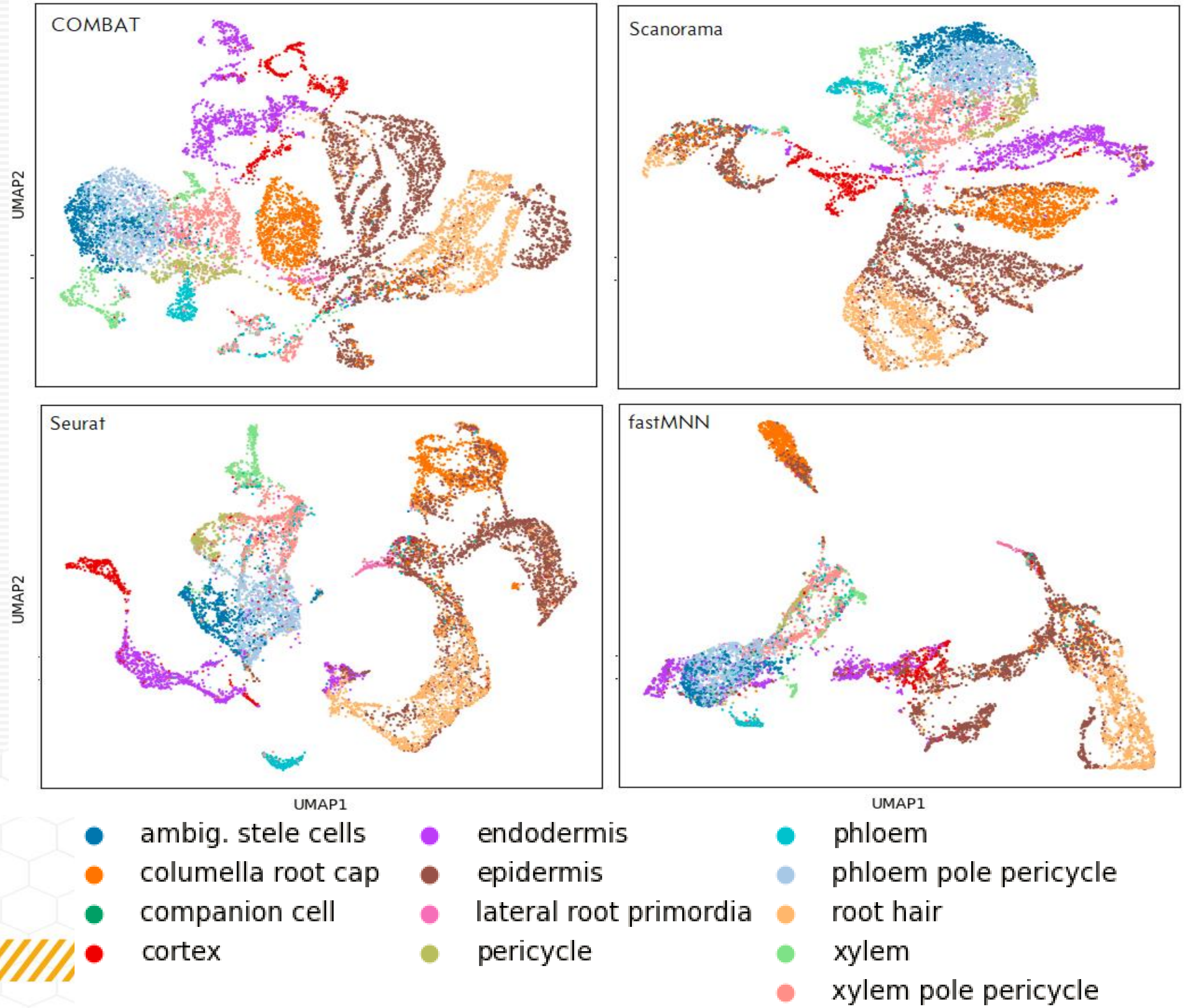
| | COMBAT | Scanorama | Seurat | fastMNN |
|---|---|---|---|---|
| NMI_cluster/label | 0.638 | 0.617 | 0.607 | 0.527 |
| ARI_cluster/label | 0.446 | 0.398 | 0.434 | 0.313 |
| ASW_label | 0.504 | 0.519 | 0.513 | 0.494 |
| ASW_label/batch | 0.925 | 0.873 | 0.878 | 0.891 |
| isolated_label (F1) | 0.222 | 0.055 | 0.074 | 0.075 |
| isolated_label (silhouette) | 0.569 | 0.502 | 0.495 | 0.545 |
| graph_conn | 0.859 | 0.895 | 0.897 | 0.926 |
| hvg_overlap | 0.194 | 0.020 | 0.085 | 0.022 |

# UMAP Plots for A. thaliana datasets

UMAP Batch

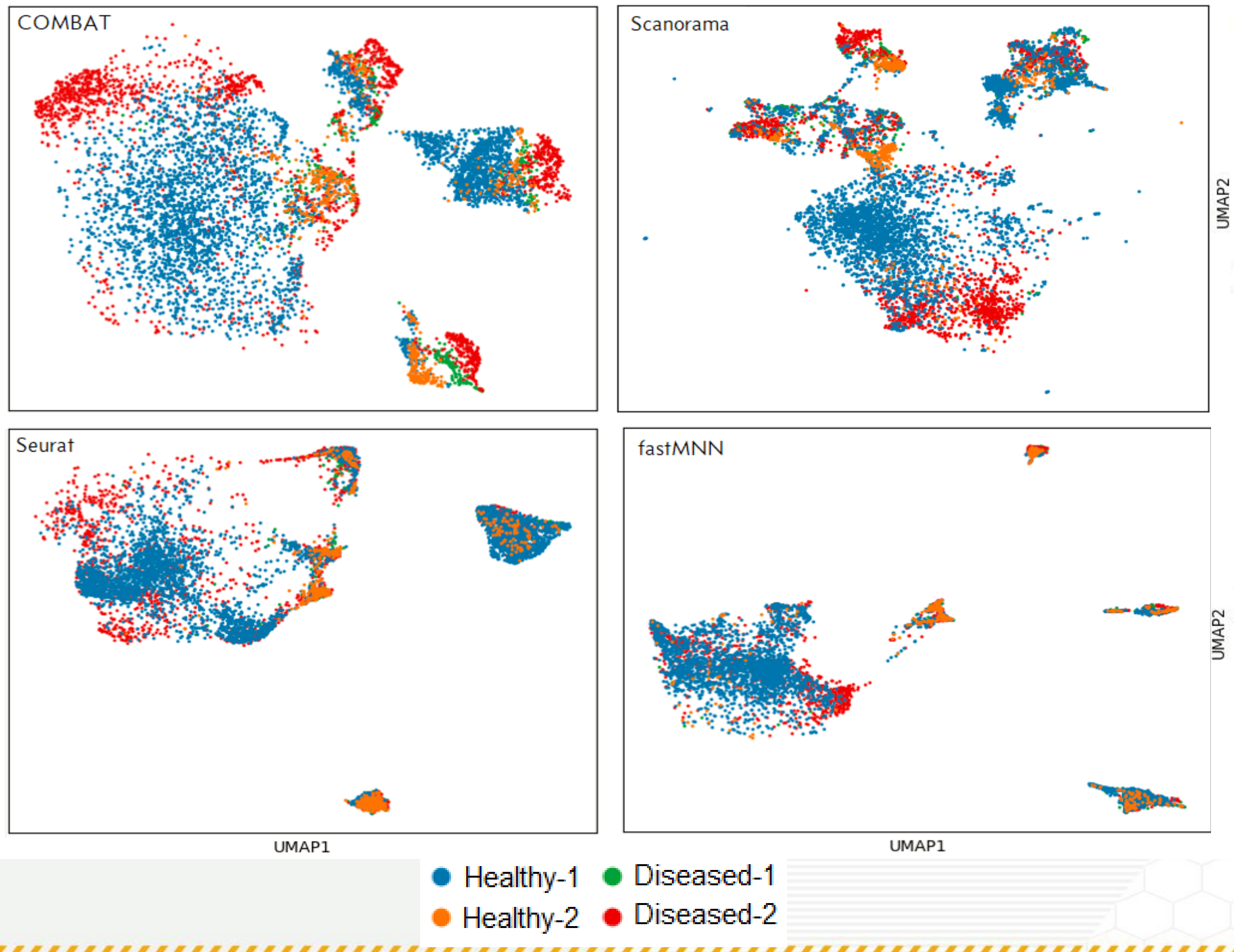



UMAP Cell Type



# UMAP Plots Aortic Valve Dataset

## UMAP Batch




## UMAP Cell Type